

**(a) the substrate = any per-token attention KV**

| attention variant              | status  |
|--------------------------------|---------|
| FlashAttn / paged / vLLM       | free    |
| GQA / MQA                      | free    |
| MLA (DeepSeek-V2/V3)           | adapter |
| M-RoPE (sectioned/interleaved) | adapter |
| Sliding-window (Gemma)         | fixed   |
| Hybrid attn+SSM (Falcon-H1)    | partial |
| Seq.-dim compress (V4 CSA/HCA) | open    |
| RWKV / Mamba / diffusion       | out     |

**(b) images are position-portable too**

